MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON,

RECTIFIER ASSEMBLY, HIGH VOLTAGE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for 500 milliampere silicon, encapsulated, uncompensated rectifier assemblies for use in high-voltage circuitry, and operated at frequencies of less than 1 kHz.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings.

Part No.	BV V(pk)	V_{RM}(wkg) V(pk)	I_O 1/ T_A = 50°C Adc	i_f (surge) T_A = 100°C I_O = 250 mAdc t_p = 8.3 ms A(pk)	Altitude mmHg
M19500/465-1	1,800	1,500	0.5	25	15
-2	3,600	3,000	0.5	25	33
-3	6,000	5,000	0.5	25	69
-4	8,300	7,500	0.5	25	439
-5	11,000	10,000	0.5	25	439
-6	13,750	12,500	0.5	25	439
-7	16,500	15,000	0.5	25	439
-8	22,000	20,000	0.5	25	439
-9	27,500	25,000	0.5	25	439

1/ Derate at .005 Adc/°C above T_A = 50°C.

OPERATING TEMPERATURE: -65° to +150°C.
STORAGE TEMPERATURE: -65° to +150°C.

2. APPLICABLE DOCUMENTS

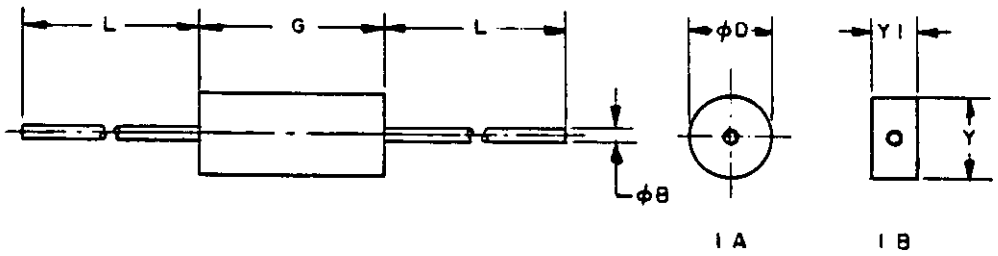
2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

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PART NO.	DIMENSIONS					
	LTR DIM.	MIN	MAX	MILLIMETERS MIN	MILLIMETERS MAX	SEE FIGURE
M19500/465-1	L	1.250	1.500	31.75	38.10	1A
	ØB	.025	.040	.64	1.02	
	G	.470	.530	11.94	13.46	
	ØD	.345	.405	8.77	10.28	
M19500/465-2	G	.970	1.030	24.64	26.16	
	ØD	.345	.405	8.77	10.29	
M19500/465-3	G	.970	1.030	24.64	26.16	
	ØD	.345	.530	8.77	13.46	
M19500/465-4 and -5	L	1.250	1.500	31.75	38.10	1B
	ØB	.040	.055	1.02	1.40	
	Y	.490	.630	12.45	16.00	
	Y_1	.240	.410	6.10	10.41	
	G	1.900	2.200	48.26	55.88	
M19500/465-6 and -7	G	2.740	3.100	69.60	78.74	
M19500/465-8 and -9	G	3.740	4.200	95.00	106.68	

FIGURE 1. Dimensions and configuration.

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STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations and symbols. The abbreviations and symbols used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. Rectifiers shall be of the design, construction, and physical dimensions specified and as shown on figure 1. The rectifier assembly shall consist of discrete controlled avalanche hermetically sealed diodes interconnected in a series string. The completed assembly of diodes and other internal structures shall be encapsulated in a plastic material which polymerizes to a rigid condition by virtue of a chemical cross-linking mechanism. The encapsulation material shall be free of voids either visible or as evidenced by failure to pass the environmental corona test specified. Each controlled avalanche junction shall withstand a minimum breakdown voltage of 650 Vdc. The supplier, at his option, may select other series junctions which have minimum breakdown voltages of 950 or 1,250 Vdc. The rectifier assembly shall not internally include or utilize component capacitors or resistors for compensation. Only those discrete diodes which have passed the tests in table I shall be used in the assembly. Discrete diodes shall be manufactured and tested by the rectifier assembly manufacturer.

3.4 Performance characteristics. The performance characteristics shall be as specified in table I for the discrete diodes and as specified in tables II, III and IV for the rectifier assemblies.

3.4.1 Process-conditioning, testing, and screening of discrete diodes. Process-conditioning, testing, and screening for discrete diodes shall be as specified in 4.5. Discrete diodes shall be devices manufactured on the same production line through final seal, by the same fabrication techniques, and to the same device design with the same material requirements.

3.5 Marking. The marking shall be placed on each device in accordance with MIL-S-19500.

3.5.1 Polarity marking. The polarity marking shall be indicated with a diode symbol or a contrasting color band to denote the cathode end. No color coding shall be permitted.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, III, and IV. Subgroups 1 and 2 of group B inspection may be performed on an inspection sublot of any type to qualify all types. Subgroups 3 and 4 of group B inspection shall be performed on an inspection sublot of the highest and lowest voltage units being qualified. Subgroup 1 of group C inspection shall be performed on an inspection sublot of the highest voltage unit of each pressure rating being qualified.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in groups A, B, and C.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table II.

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4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table III. Subgroups 1 and 2 may be performed on a lot basis. Subgroups 3 and 4 shall be performed on a sublot basis.

4.3.3 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table IV. This inspection shall be conducted on the initial lot, and thereafter every 6 months during production. All tests shall be performed on a sublot basis or on a sublot of the highest voltage type for each altitude rating.

4.3.4 100% inspection. 100 percent inspection shall consist of the examinations and tests specified in subgroup 1 of table I.

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, III, IV and as follows.

4.4.1 Steady-state operation life. This test shall be conducted with a half-sine waveform of the specified peak voltage impressed across the rectifier in the reverse direction followed by a half-sine waveform of the specified average rectifier current. The forward conduction angle of the rectified current shall not be greater than 180° nor less than 150°. The temperature differentials from lead to mount and from mount to ambient shall not exceed 5°C with lead connections made at a minimum distance from the body of device of 3/8 inch. This test shall be conducted with no forced air flow across the body of the assembly.

4.4.2 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.4.3 Corona test. The method of corona detection, measurement and display shall conform to the following requirements. The rectifier terminals shall be connected together and connected to the high potential side of the corona test equipment shown in figure 2. A test voltage of 25 kV rms at 60 Hz shall be applied between the terminals and the ground plane which shall be no more than two inches from the major diameter of the rectifier assembly. The test potential shall be applied for not less than 20 seconds. The corona tester shall be a Biddle Company No. 661409 AC Corona Tester or equivalent. The corona discharge as viewed on the tester display screen shall show no more than 50 discharges per cycle (whole ellipse). None of the 50 discharges shall exceed 5 picocoulombs. In order to minimize the masking of internal corona by any external corona, rounded terminal hardware of suitable design may be added to the connectors during the corona test. In addition, the test may be performed with the unit either entirely or partially immersed in a suitable dielectric fluid in order to reduce external corona.

4.4.4 Resistance to solvents. The rectifier assemblies shall be subjected to tests in accordance with method 215 of MIL-STD-202. The following details shall apply:

(a) All areas of the encapsulated assembly where marking has been applied shall be brushed.
(b) After subjection to the tests there shall be no evidence of mechanical damage to the device and markings shall have remained legible.

4.5 Process-conditioning, testing, and screening of discrete diodes. The procedure for process-conditioning and testing shall be in accordance with 4.5.1 through 4.5.6.3, table I, and figures 3 and 4. Process-conditioning shall be conducted on 100 percent of the discrete diode lot prior to assembly and encapsulation of the high-voltage rectifiers.

4.5.1 High-temperature storage. All devices shall be stored for at least 48 hours at a minimum temperature (T_A) of 200°C.

4.5.2 Thermal shock (temperature cycling). All devices shall be subjected to a thermal shock (temperature cycling) in accordance with MIL-STD-750, method 1051, test condition C, except that 10 cycles shall be continuously performed. All devices shall be maintained at each end temperature until thermal equilibrium has been reached but not less than 15 minutes.

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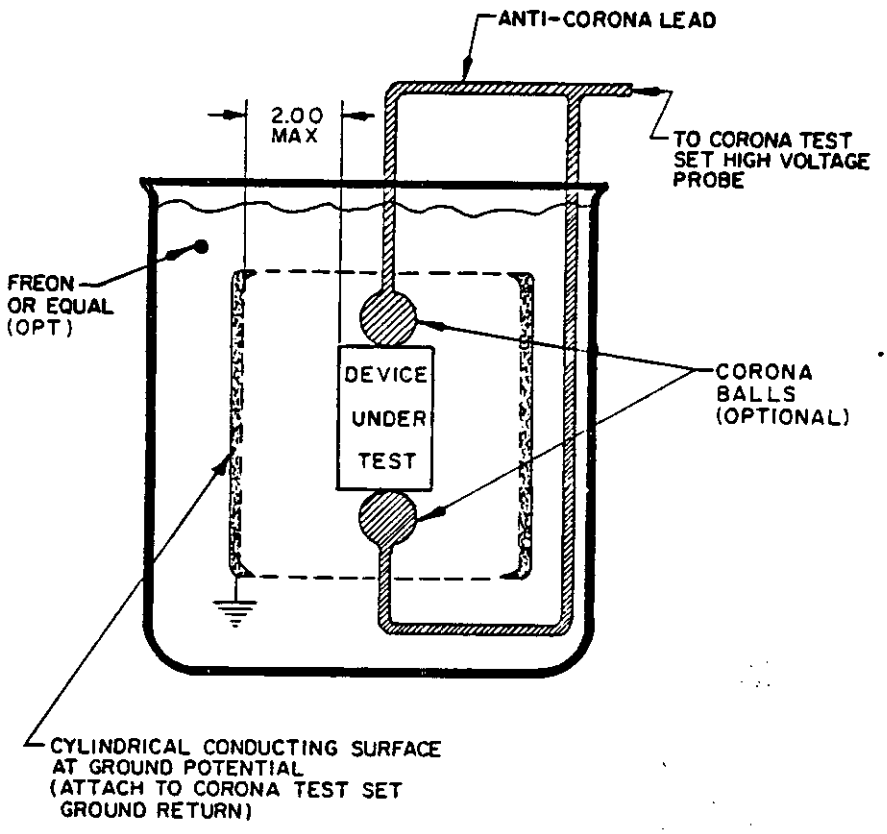


FIGURE 2. Corona test procedure.

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4.5.3 Peak transient reverse power. The circuit of figure 3 or a similar circuit, at the option of the supplier, shall be used to apply a single cycle reverse power surge to discrete diodes or to rectifier assemblies. The reverse power surge used may be applied in the form of a rectangular pulse of 18 to 24 microseconds duration or as a one-half sinusoidal pulse of 8.3 milliseconds duration.

4.5.4 Avalanche characteristics. The breakdown characteristics of each device shall be graphically viewed on an oscilloscope with a peak reverse current, i_r, of 100 μA minimum. The scope voltage sensitivity shall be 100 V/cm and current sensitivity shall be 20 μA/cm. For assemblies the horizontal sensitivity shall be 1/5 V_{RM}(wkg)/cm. All devices when viewed on the oscilloscope shall present sharp knee characteristics. Any discontinuity or dynamic instability of the trace shall be cause for rejection. If heating effects are present, only those devices exhibiting a positive temperature coefficient shall be accepted.

4.5.5 Reverse-recovery time. Recovery time of each device shall be measured by the circuit shown on figure 4 or equivalent method. The pulse generator shall have a pulse repetition frequency of 1 kHz maximum and a pulse width of 10 μs minimum. Recovery conditions: 0.5 ampere forward current to 1.0 ampere reverse current. Recovery time is measured from t = 0 to when rectifier recovers to 0.25 ampere.

TABLE I. Process-conditioning, testing and screening of discrete diodes.

Examination or test	MIL-STD-750		L_T P_D	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1							
High temperature storage		See 4.5.1	100% test	---	---	---	---
Thermal shock (temperature cycling)		See 4.5.2		---	---	---	---
Peak transient reverse power		See 4.5.3 10 m J for 20 μs rectangular pulse (or 100 m J for 8.3 ms sinusoidal pulse)		---	---	---	---
Avalanche characteristics		See 4.5.4		V_R	650	1,400	Vdc
Subgroup 2			3				
Forward voltage	4011	i_f = 1.5 A(pk) 250 ms pulse width $\leq$ 2% duty cycle		v_f	---	1.1	V(pk)
Reverse current	4016	DC method 650 $\leq V_R \geq$ 1,250		I_R	---	1.0	μAdc
Subgroup 3			5				
Reverse recovery time		See 4.5.5		t_{rr}	---	1.5	μs
Subgroup 4			5				
Hermetic seal test	1071	Test condition G or H for fine leaks; test condition D or E for gross leaks		---	---	---	---

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TABLE II. Group A inspection.

Examination or test	MIL-STD-750		L_{T_P}		Limits		
	Method	Details	D	Symbol	Min	Max	Unit
Subgroup 1			10				
Visual and mechanical examination	2071			---	---	---	---
Subgroup 2			5				
Forward voltage	4011	i_f = 1.5 A(pk) Pulse width = 250 ms duty cycle ≤ 2%					
M19500/465-1				V_f	0.8	3.6	V(pk)
-2				V_f	2.4	6.6	V(pk)
-3				V_f	4.0	9.9	V(pk)
-4				V_f	6.5	15	V(pk)
-5				V_f	8.5	19	V(pk)
-6				V_f	11.0	23	V(pk)
-7				V_f	13.0	27	V(pk)
-8				V_f	17.0	37	V(pk)
-9				V_f	22.0	45	V(pk)
Reverse current	4016	DC method $V_R = V_{RM}$(wkg) (see 1.3)		I_R	---	1.0	μAdc
Breakdown voltage	4021	I_R = 100 μAdc max		BV	See 1.3	---	V(pk)
Subgroup 3			10				
High temperature operation:		T_A = 100°C					
Reverse current	4016	DC method $V_R = V_{RM}$(wkg) (see 1.3)		I_R	---	100	μAdc
Subgroup 4			5				
Peak transient reverse power	---	See 4.5.3					
M19500/465-1				---	0.2	---	Joules
-2				---	0.5	---	Joules
-3				---	0.8	---	Joules
-4				---	1.2	---	Joules
-5				---	1.6	---	Joules
-6				---	2.0	---	Joules
-7				---	2.4	---	Joules
-8				---	3.2	---	Joules
-9				---	4.0	---	Joules
Subgroup 5			3				
Avalanche characteristic	---	See 4.5.4		---	---	---	---
Subgroup 6			10				
Corona test (applicable to types 7.5 kV through 25 kV)	---	See 4.4.3		---	---	---	---

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TABLE III. Group B inspection.

Examination or test	MIL-STD-750		$L_{T_P D}$	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			10				
Physical dimensions	2066	See figure 1		---	---	---	---
Subgroup 2			20				
Solderability	2026	Immersion depth to within 0.10 inch of body		---	---	---	---
Thermal shock (temperature cycling)	1051	Test condition C; T (high) = 150°C; 50 cycles minimum; holding time of 5 minutes at temperature extremes		---	---	---	---
Terminal strength:	2036						
Tension		Test condition A; 6 lbs; t = 30 sec		---	---	---	---
Lead fatigue		Test condition E; 1 lb		---	---	---	---
Moisture resistance	1021	Omit initial conditioning		---	---	---	---
End points:							
(Same as subgroup 2 of group A)							
Subgroup 3			10				
Forward-surge current	---	Superimpose 10 surges, 1 surge per minute; 25 A(pk); t_p = 8.3 ms; I_O = 250 mAdc; T_A = 100°C		---	---	---	---
End points:							
(Same as subgroup 2 of group A)							
Subgroup 4			λ = 10				
High-temperature life (nonoperating)	1031	T_A = 150°C		---	---	---	---
End points:							
(Same as subgroup 2 of group A)							
Subgroup 5			λ = 10				
Steady state operation life	1026	I_O = 550 mAdc T_A = 25°C; f = 60 Hz: V_R(pk) = V_{RM}(wkg) (see 1.3)		---	---	---	---

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TABLE III. Group B inspection. - Continued

Examination or test	MIL-STD-750		$L_{T_{P_D}}$	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 5 - Continued							
End points:							
(Same as subgroup 2 of group A)							

TABLE IV. Group C inspection.

Examination or test	MIL-STD-750		$L_{T_{P_D}}$	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			20				
Barometric pressure, reduced (altitude operation)	1001	For pressure (alt) see 1.3; t = 60 sec		---	---	---	---
Measurement during test:							
Reverse current	4016	DC method $V_R = V_{RM}$(wkg) See 1.3		I_R	---	1.0	µAdc
Salt atmosphere (corrosion)	1041			---	---	---	---
Subgroup 2			20				
Shock	2016	Nonoperating: 1,500 G; t = 0.5 ms, 5 blows in each orientation X_1, Y_1, and Y_2		---	---	---	---
Vibration, variable frequency	2056	Nonoperating; 25 G		---	---	---	---
End points:							
(Same as subgroup 1 of group A)							
Subgroup 3			10				
Low-temperature operation:		T_A = -65°C					
Forward voltage	4011	i_f = 1.5 A(pk) (pulsed)					
M19500/465-1				V_f	---	3.7	V(pk)
-2				V_f	---	8.6	V(pk)
-3				V_f	---	12.8	V(pk)
-4				V_f	---	19.5	V(pk)
-5				V_f	---	24.7	V(pk)
-6				V_f	---	29.9	V(pk)
-7				V_f	---	35.1	V(pk)
-8				V_f	---	48.1	V(pk)
-9				V_f	---	58.5	V(pk)
Breakdown voltage	4021	V_{RM}(wkg) - See 1.3		I_R	---	100	µAdc

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TABLE IV. Group C inspection. - Continued

Examination or test	MIL-STD-750		L_{T_P} D	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 4			10				
Flammability	---	MIL-STD-202, method 111; allowable burning time = 5 sec. Point of impingement shall be at midpoint of diode body		---	---	---	---
Subgroup 5			10				
Resistance to solvents	215	See 4.4.4		---	---	---	---

4.5.6 Preburn-in tests. The parameters I_R and v_f of table I shall be measured and the data recorded for all devices in the lot. All devices shall be handled or identified such that the delta end points can be determined after the burn-in test. All devices which fail to meet these requirements shall be removed from the lot. Quantity removed shall be noted on the lot history.

4.5.6.1 Burn-in test. All devices shall be operated for 96 hours minimum under the following conditions:

$T_A = 100°C$ $I_O = .25$ Adc

$$V_R = \frac{V_{RM(wkg)}}{\text{Number of discrete devices in assembly}}$$

Mounting conditions per method 1026 of MIL-STD-750.

4.5.6.2 Post burn-in test. The parameters I_R and v_f of table I shall be retested after burn in and the data recorded for all devices in the lot. The parameters measured shall not have changed during the burn-in test from the initial value by more than the specified amount as follows:

ΔI_R = 100 percent of the initial value or 0.5 μAdc, whichever is greater.

$\Delta v_f = \pm 0.05$ V(pk) from the initial value.

4.5.6.3 Burn-in test failures (screening). All devices that exceed the delta limits of 4.5.6.2 or the limits of those tests performed on table I shall be removed from the inspection lot, and the quantity removed shall be noted on the lot history. Where the quantity removed exceeds 10 percent of the total inspection lot on burn-in test, the entire lot shall be unacceptable as discrete diodes for the rectifier assembly, and these discrete diodes shall not be resubmitted as another lot.

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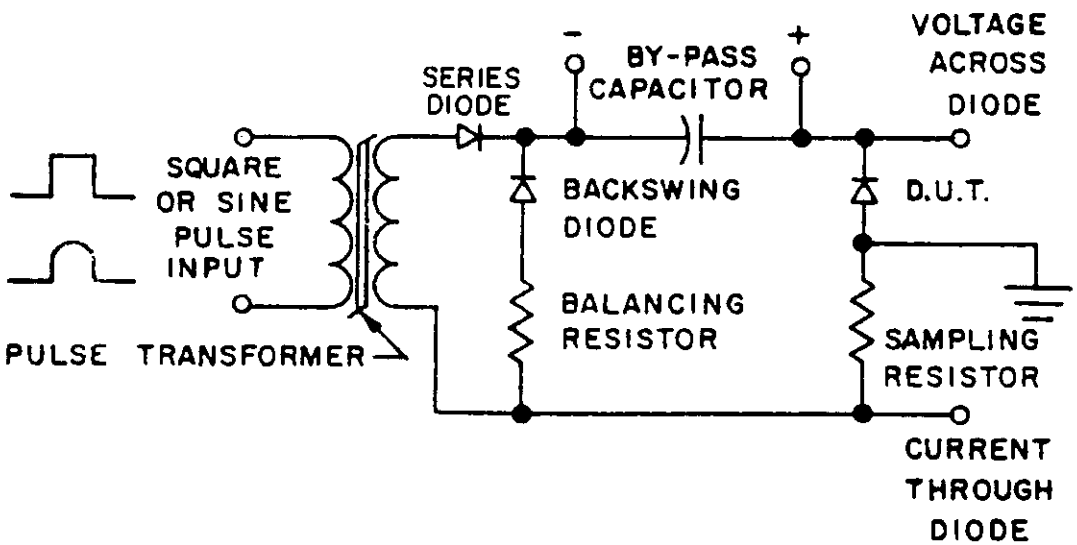
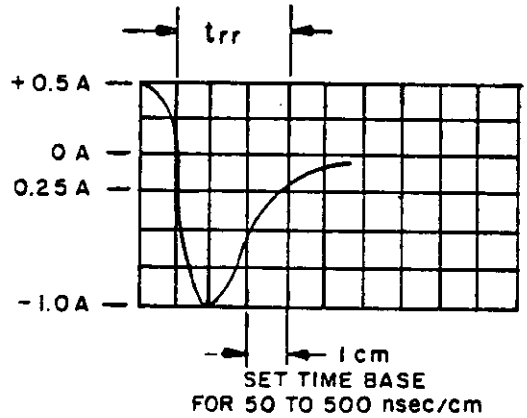
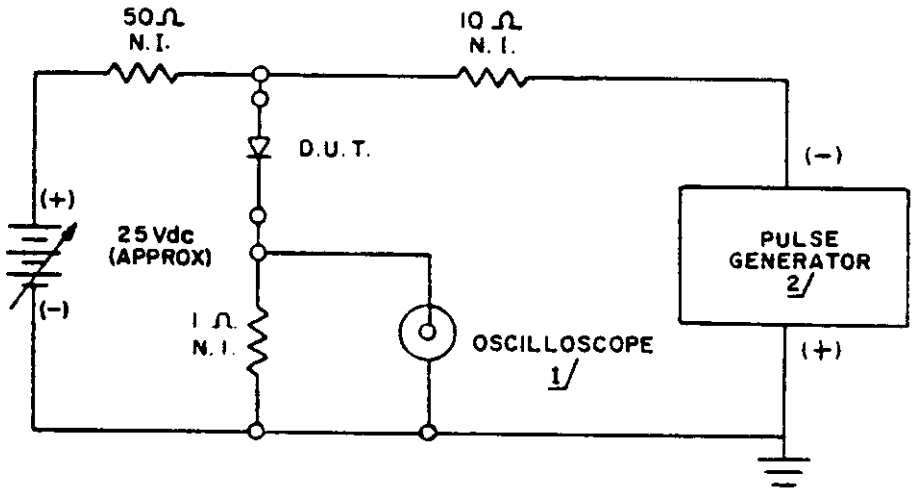


FIGURE 3. Peak transient reverse power test circuit.

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NOTES:
1/ Rise time = 7 nsec maximum, input impedance = 1 megohm, 22 pf.
2/ Rise time = 10 nsec maximum, source impedance = 50 ohm.

FIGURE 4. Reverse recovery trace characteristic and test circuit diagram.

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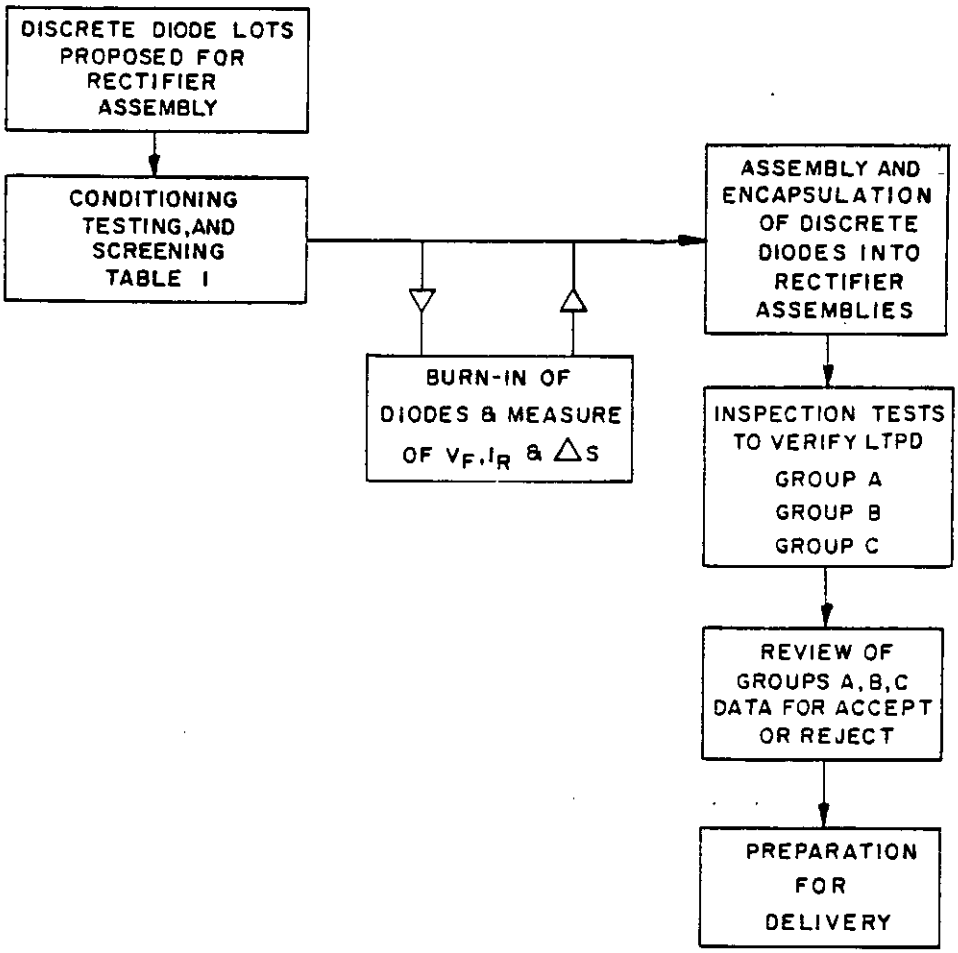


FIGURE 5. Procedure diagram.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery. Preparation for delivery shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

Custodians:
Army - EL
Navy - EC
Air Force - 17

Review activities:
Army - MU, MI
Air Force - 11, 70, 80
DSA - ES

User activities:
Army - SM
Navy - MC, CG, AS, OS
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

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